2024 Spring 生化学 2 English, Home work#1(250310)　　dead line: 23:59, March 17, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 20 分）

(1) Enzymes are all protein (　　)
(2) If you change the Enzyme concentration, K_m value is also changed (　　)
(3) Enzyme reduces the activation energy $\Delta G^{\ddagger}$ (　　)
(4) Enzyme optimizes to bind to transition state (　　)
(5) Enzymatic reaction can be fully explained by “lock and key” model (　　)
(6) If the substrate concentration [S] is much lower than the K_m value, the product production speed is linearly increased with [S] (　　)
(7) With same enzyme, k_{cat} is constant for different substrate (　　)
(8) In Double-Reciprocal Plot (Lineweaver-Burk equation), the slope is K_m/V_{max} (　　)
(9) In the ping-pong mechanism, ternary complex (ES_1S_2) is made as intermediate (　　)
(10) In the uncompetitive inhibition, inhibitor binds to the same place to that of substrate (　　)
(11) In the transition state of trypsin, tetrahedral acyl is stabilized by hydrogen-bonds from Ser195 and Gly193 (　　)
(12) In some enzyme, metal ions contribute the enzymatic reaction (　　)
(13) Allosteric enzyme follows the Michaelis-Menten equation (　　)
(14) At low [S], allosteric enzyme take R (active) state (　　)
(15) In Ubiquitination, ubiquitin added lipid to proteins as a tag (　　)
(16) Phosphorylation is reversible process (　　)
(17) Phosphorylation adds positive charge to the target protein (　　)
(18) Phosphatase uses ATP (　　)
(19) Kinase randomly phosporylate the target (　　)
(20) Chymotrypsin needs proteolytic cleavage for activation (　　)

Q2. Please develop the expression for Michaelis-Menten equation. If you use some assumption, please also clearly state the assumption (10 分)

Q3. Please draw Double-Reciprocal Plot (Lineweaver-Burk equation), and calculate the K_m

and V_{max} values using the following data set. Using excel etc is ok (各 5 分、共 10 分)

[S] mM	Vo μM/min
0.1	0.48
0.2	0.91
0.5	2
1	3.3
2	5
4	6.7
10	8.3

Q4. Please draw Double-Reciprocal Plot (Lineweaver-Burk equation, 4 分), determine the inhibition mechanism (3 分), and estimate the K_i value (3 分)

	Vo μM/min			
[S] mM	[I] = 0mM	[I] = 1mM	[I] = 2mM	[I] = 10mM
0	0	0	0	0
0.1	0.48	0.32	0.244	0.0826
0.2	0.91	0.63	0.476	0.164
0.5	2	1.4	1.11	0.4
1	3.3	2.5	2	0.769
2	5	4	3.3	1.428
4	6.7	5.7	5	2.5
10	8.3	7.7	7.1	4.54